HARSH MAHTO

B.Tech — Computer Science & Engineering | 3rd Year

Shri Shankaracharya Technical Campus, Bhilai, Chhattisgarh

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CAREER OBJECTIVE

Motivated third-year Computer Science and Engineering student with Java as my primary programming language and knowledge of C, C++, and Python. Interested in software development, problem-solving, AI-enabled applications, and hackathon projects. Runner-up in a college GDG Hackathon, with experience working on practical technology solutions and collaborative development.

EDUCATION

Shri Shankaracharya Technical Campus, Bhilai, Chhattisgarh
Bachelor of Technology (B.Tech) — Computer Science & Engineering
Expected Graduation: 2028 | CGPA: 8.0/10

TECHNICAL SKILLS

Programming Languages: Java, C, C++, Python
Core Skills: Data Structures & Algorithms, Object-Oriented Programming, Problem Solving
Web & Development: HTML, CSS, JavaScript, FastAPI
AI / Data & APIs: ConvLSTM, TensorFlow/Keras, LLaMA 3.3-70B, Groq API
Tools & Technologies: Git, GitHub, VS Code, Leaflet.js, Chart.js, NetworkX

PROJECTS

FloodVision-Pro — AI-Powered Ward-Level Urban Flood Intelligence Platform

- Developed as a team project at SSTC to provide ward-level urban flood risk intelligence for Indian cities.
- Integrated ConvLSTM-based flood prediction with rainfall, elevation, drainage, water-proximity, and soil data for 3-day and 7-day risk analysis.
- Implemented WMLC hotspot analysis, Drain Health Index, flood-aware Dijkstra route planning, budget optimization, and Digital Twin simulation.
- Built REST APIs using Python, FastAPI and Uvicorn, with Leaflet.js and Chart.js for interactive maps and analytical dashboards.
- Integrated LLaMA 3.3-70B through the Groq API for a bilingual English/Hindi flood-management chatbot.
- Project report records 87.4% validation accuracy for the ConvLSTM model and sub-500ms API performance under normal conditions.

Smart Student Activity Detection System

Developed an IoT- and AI-based system using ESP32-CAM, YOLOv8 object detection, and YOLOv8-pose to monitor and classify student study activities in real time.

Built a Python Flask backend with OpenCV, PySerial, JSON logging, and a responsive HTML/CSS/JavaScript dashboard for activity tracking, analytics, and continuous monitoring.

Achieved 450ms average frame-processing latency, 89.3% object-detection accuracy, and 87.6% pose-estimation accuracy under reported test conditions.

Legal Aid AI Assistant

- Developed a web-based AI assistant concept aimed at helping users understand legal issues and access relevant legal information.
- Worked on frontend and backend integration with a focus on a simple and accessible user interface.

Smart Crop Monitoring System

- Worked on a technology-based solution for monitoring crop conditions and supporting agricultural decision-making.
- Developed as part of hackathon-oriented project work.

ACHIEVEMENTS

- Runner-Up — GDG College Hackathon, Shri Shankaracharya Technical Campus

CERTIFICATIONS & TRAINING

- Infosys — Programming using Java, successfully completed June 16, 2026.
- Java Vocational Training — Internshala, successfully completed.

INTERESTS

Software Development • Java Development • Data Structures & Algorithms • AI Applications • Hackathons • Problem Solving